# CSCI 441 - Software Engineering

# SuperAsk.AI: AI-Powered Government Service Assistant

Report One, Part Two

Team AO3

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Project URL: https://github.com/Monunhh/Super-ask.git

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**Individual Contributions Breakdown**

| Team Member | Contribution to Report One, Part Two |
|---|---|
| Yean Fong | Contributed to writing Section 3b: Actors and Goals; prepared Section 5: Project Management, Gantt chart, team responsibilities, and references. |
| Doungkhe Neang | Contributed to writing Section 3b: Actors and Goals. |
| Sereyprophea Sarin | Contributed to writing Section 3a: Stakeholders. |
| Phallamonunhh Chun | Contributed to writing Section 3c-i: Casual Descriptions; Section 3c-ii: Use Case Diagram; Section 3c-iv: Fully-Dressed Descriptions |
| Bunkheang Heng | Contributed to writing Section 3c-iii: Traceability Matrix; Section 3d: System Sequence Diagrams |
| Heejae Choi | Contributed to writing Section 4a: Preliminary design |

## 3. Functional Requirement Specification and Use Cases

### a. Stakeholders

Having a reliable, centralized way to access accurate government service information can reduce the confusion, wasted trips, and missed deadlines that currently push citizens toward costly agencies or repeated failed visits to the wrong office. This can also ease pressure on government offices themselves, since fewer citizens will arrive with incomplete documents or outdated information. Making government procedures easier to understand and follow benefits individual citizens as well as the broader public institutions they depend on. However, we have identified these primary stakeholders:

**Citizens (Individual Users):** use the system to look up information about government procedures such as document notarization, birth registration, and voter registration. These users will benefit by having accurate, up-to-date requirements, deadlines, and fees available in one place instead of scattered across ministry websites, Facebook posts, and word of mouth. Having a bilingual chat interface that clearly cites its sources will allow citizens to prepare correctly before visiting an office, reducing the stress, wasted time, and unexpected fees described in Scenarios 1 through 3.

**Government Officers and Public Institutions:** Do not use SuperAsk.AI directly, but they publish the official information it relies on and remain the point of contact for citizens who prepare through the app. These stakeholders will benefit by receiving fewer repetitive questions and fewer citizens arriving with missing or incorrectly formatted documents. Reducing this repetitive back-and-forth, as described in Scenario 4, will free up staff time to process more cases and reduce the strain caused by understaffing and constant interruptions.

**Ministries and Government Agencies:** These are the organizations whose official procedures, requirements, and contact information are published on their websites, Facebook pages, and Telegram channels, and collected into the system from there. These stakeholders have an interest in ensuring the information SuperAsk.AI displays is accurate, properly sourced, and kept current, since any errors would reflect on the ministry itself. As more ministries are added under REQ-30, this group will continue to grow alongside the system's coverage.

**Professor Hieu Vu and the CSCI 441 Course** serve as the academic sponsor and evaluator of the project. This stakeholder has an interest in seeing the team apply sound software engineering practices throughout requirements gathering, design, implementation, and testing, and in reviewing the team's progress at each report and demonstration milestone.

**Team AO3 (Developers):** are deeply invested in the success of SuperAsk.AI. Each member is responsible for a specific area, including AI and retrieval accuracy, backend and database development, content curation, quality assurance, and interface design. The team's dedication to building a functional, trustworthy, and easy to use system is critical to ensuring the project meets both its academic requirements and its intended real world purpose.

b. **Actors and Goals**

| **Actor** | **Participating / Initiating** | **Role** | **Goal** |
|---|---|---|---|
| Citizen / User | Initiating | The User interacts with SuperAsk.AI to ask questions about Cambodian government procedures and services. | The goal of the User is to receive accurate government-service information in Khmer or English, including procedures, required documents, fees, deadlines, office information, and verified sources. |
| AI Model Service | Participating | The AI Model Service processes the user's question together with retrieved government information to generate a relevant response. | The goal of the AI Model Service is to generate a clear response based on the approved information provided by SuperAsk.AI. |
| Telegram Platform | Participating | The Telegram Platform provides an additional communication channel through which users may interact with SuperAsk.AI. | The goal of the Telegram Platform is to transmit user questions to SuperAsk.AI and deliver generated responses back to the user. |
| Messenger Platform | Participating | The Messenger Platform provides an additional communication channel through which users may interact with SuperAsk.AI. | The goal of the Messenger Platform is to transmit user questions to SuperAsk.AI and deliver generated responses back to the user. |
| Scheduler (System Timer) | Initiating | The Scheduler starts SuperAsk.AI's content refresh at set intervals, without any user action. | The goal of the Scheduler is to keep the stored government information current, so that answers and last-verified dates reflect what ministries have most recently published. |
| Official Sources (Ministry Websites, Facebook Pages, Telegram Channels) | Participating | The Official Sources are the public online channels where ministries publish procedures, fees, deadlines, and contact details. SuperAsk.AI crawls them to collect its content. | The goal of the Official Sources is to provide the up-to-date official information that every SuperAsk.AI answer and citation is based on. |

## c. Use Cases

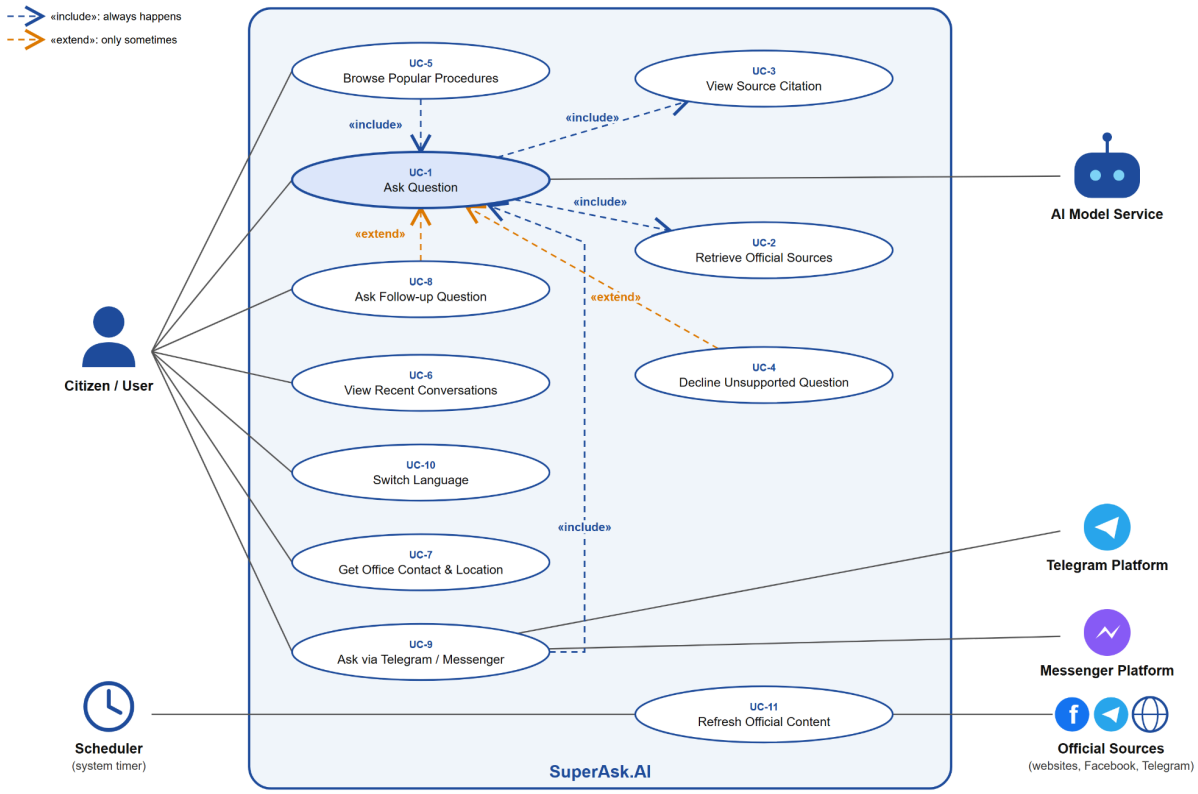
### 3c-i. Casual Description

The use cases below cover every system requirement from Report One, Part One. Each one lists the requirements it responds to. UC-1 to UC-10 are started by the Citizen / User. UC-11 is started by the system's own scheduler. It keeps the stored government information current by crawling official ministry websites, Facebook pages, and Telegram channels.

| Use Case | Name | Casual Description | Requirements |
|---|---|---|---|
| UC-1 | AskQuestion | The user types a question about a Cambodian government service in Khmer, English, or a mix of both, and submits it. SuperAsk.AI finds the relevant official information and returns an answer. Where the source has them, the answer lists the procedure steps in order, the required documents, fees and late fees, deadlines or validity periods, and office hours. | REQ-1, REQ-2, REQ-3, REQ-5, REQ-16, REQ-17, REQ-18, REQ-19, REQ-20, REQ-31, REQ-32, REQ-33 |
| UC-2 | RetrieveOfficialSources | The system searches its stored official government content for the passages most relevant to the question and passes only those to the AI Model Service. It is always included in UC-1. | REQ-4, REQ-5, REQ-8 |
| UC-3 | ViewSourceCitation | The user sees the official source behind the answer: the ministry, the document or page, a link, the effective date, and the last-verified date. The citation is kept visually separate from the answer text. It is always included when a sourced answer is shown. | REQ-6, REQ-28, REQ-34 |
| UC-4 | DeclineUnsupportedQuestion | If no verified official source covers the question, or the question is out of scope, the system says it cannot answer. It does not guess. It tells the user which areas it does cover so they can rephrase. This extends UC-1. | REQ-7 |
| UC-5 | BrowsePopularProcedures | The user picks a common procedure, such as passport application or birth certificate, from the sidebar or the welcome screen. The system sends it as a question and answers it as in UC-1. | REQ-9, REQ-35 |

| | | | |
|---|---|---|---|
| UC-6 | ViewRecentConversations | The user opens the list of recent questions and conversations and selects one to go back to a topic they viewed earlier. | REQ-10, REQ-36 |
| UC-7 | GetOfficeContactAndLocation | The user asks for the contact details and location of the ministry or office that handles their procedure. The system shows the phone number, address, office hours, and a map link. If there is no online contact, it says so and shows the office location instead. | REQ-11, REQ-12, REQ-13, REQ-20, REQ-38, REQ-39 |
| UC-8 | AskFollowUpQuestion | After receiving an answer, the user asks a follow-up question, such as "how much does it cost?". The system uses the earlier conversation to work out what "it" refers to. This extends UC-1. | REQ-14 |
| UC-9 | AskThroughMessagingChannel | The user sends a question to SuperAsk.AI through Telegram or Messenger instead of the website. The platform forwards it, the system answers it through UC-1, and the platform sends back the answer with its source. | REQ-15 |
| UC-10 | SwitchLanguage | The user switches the interface between Khmer and English. Answers come back in the language the question was asked in. | REQ-2, REQ-37 |
| UC-11 | RefreshOfficialContent | On a schedule, the system crawls the registered official sources: ministry websites, Facebook pages, and Telegram channels. It saves new or changed content with its source link and fetch date, and updates the stored information and its last-verified date. New ministries are added by registering their sources, without changing the answer-generation logic. | REQ-4, REQ-8, REQ-29, REQ-30 |

### 3c-ii. Use Case Diagram




The diagram above shows how the SuperAsk.AI use cases relate to each other and to the actors. AskQuestion (UC-1) is the central use case: it always includes RetrieveOfficialSources (UC-2) and ViewSourceCitation (UC-3), because every answer is looked up and cited. BrowsePopularProcedures (UC-5) and AskThroughMessagingChannel (UC-9) include AskQuestion, since they are other ways of asking the same question. AskFollowUpQuestion (UC-8) and DeclineUnsupportedQuestion (UC-4) extend AskQuestion, because they only happen in some conversations. RefreshOfficialContent (UC-11) is started by the Scheduler and collects content from the official ministry websites, Facebook pages, and Telegram channels.

### 3c-iii. Traceability Matrix (1) - System Requirements to Use Cases

The traceability matrix below maps each system requirement from Report One, Part One to the use cases that satisfy it. The priority weight of a use case is the sum of the priority weights (PW) of the requirements it covers. Nonfunctional requirements REQ-21 to

REQ-27 (usability, accuracy, refusal rate, response time, screen width, Khmer display, and security) are quality constraints on the whole system rather than on a single use case, so they are verified through system testing instead of being traced here.

| Req't | PW | UC-1 | UC-2 | UC-3 | UC-4 | UC-5 | UC-6 | UC-7 | UC-8 | UC-9 | UC-10 | UC-11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| REQ-1 | 5 | X | | | | | | | | | | |
| REQ-2 | 5 | X | | | | | | | | | X | |
| REQ-3 | 4 | X | | | | | | | | | | |
| REQ-4 | 5 | | X | | | | | | | | | X |
| REQ-5 | 5 | X | X | | | | | | | | | |
| REQ-6 | 5 | | | X | | | | | | | | |
| REQ-7 | 5 | | | | X | | | | | | | |
| REQ-8 | 5 | | X | | | | | | | | | X |
| REQ-9 | 4 | | | | | X | | | | | | |
| REQ-10 | 3 | | | | | | X | | | | | |
| REQ-11 | 4 | | | | | | | X | | | | |
| REQ-12 | 4 | | | | | | | X | | | | |
| REQ-13 | 4 | | | | | | | X | | | | |
| REQ-14 | 4 | | | | | | | | X | | | |
| REQ-15 | 3 | | | | | | | | | X | | |
| REQ-16 | 5 | X | | | | | | | | | | |
| REQ-17 | 5 | X | | | | | | | | | | |
| REQ-18 | 5 | X | | | | | | | | | | |
| REQ-19 | 5 | X | | | | | | | | | | |
| REQ-20 | 4 | X | | | | | | X | | | | |
| REQ-28 | 4 | | | X | | | | | | | | |
| REQ-29 | 4 | | | | | | | | | | | X |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| REQ-30 | 4 | | | | | | | | | | | X |
| REQ-31 | 5 | X | | | | | | | | | | |
| REQ-32 | 5 | X | | | | | | | | | | |
| REQ-33 | 5 | X | | | | | | | | | | |
| REQ-34 | 5 | | | X | | | | | | | | |
| REQ-35 | 4 | | | | | X | | | | | | |
| REQ-36 | 4 | | | | | | X | | | | | |
| REQ-37 | 4 | | | | | | | | | | X | |
| REQ-38 | 4 | | | | | | | X | | | | |
| REQ-39 | 4 | | | | | | | X | | | | |
| **Max PW** | | **58** | **15** | **14** | **5** | **8** | **7** | **24** | **4** | **3** | **9** | **18** |

Ranked by priority weight: UC-1 (58), UC-7 (24), UC-11 (18), UC-2 (15), UC-3 (14), UC-10 (9), UC-5 (8), UC-6 (7), UC-4 (5), UC-8 (4), UC-9 (3). The five highest-weighted use cases, UC-1 AskQuestion, UC-7 GetOfficeContactAndLocation, UC-11 RefreshOfficialContent, UC-2 RetrieveOfficialSources, and UC-3 ViewSourceCitation, are elaborated in Sections 3c-iv and 3d and planned for the first demo. UC-4 DeclineUnsupportedQuestion has a low weight because it traces to a single requirement, but it will also be shown in the first demo, since it runs inside UC-1 whenever no verified source is found.

### 3c-iv. Fully-Dressed Description

The five use cases below have the highest priority weights in the traceability matrix (Section 3c-iii) and form the core of the first demonstration. In each flow, → marks an action by an actor and ← marks the system's response.

| **Use Case UC-1: AskQuestion** | |
|---|---|
| **Requirements** | REQ-1, REQ-2, REQ-3, REQ-5, REQ-16, REQ-17, REQ-18, REQ-19, REQ-20, REQ-31, REQ-32, REQ-33 |

| | |
|---|---|
| **Initiating Actor** | Citizen / User |
| **Actor's Goal** | To get accurate information about a government procedure (steps, required documents, fees, deadlines, office hours) in Khmer or English. |
| **Participating Actors** | AI Model Service |
| **Preconditions** | The SuperAsk.AI web interface is open. The stored official content has been loaded (UC-11 has run at least once). |
| **Postconditions** | The user has an answer supported by a displayed official source, or a clear statement that no verified source is available (UC-4). The question and answer appear in the recent conversations list. |
| **Flow of Events for Main Success Scenario** | → 1. **User** types a question in Khmer, English, or a mix of both into the chat input field.<br><br>→ 2. **User** submits the question by pressing Enter or the send button.<br><br>← 3. **System** shows the question in the conversation and a loading indicator.<br><br>← 4. **System** detects the question's language and normalises the Khmer text.<br><br>← 5. **System** checks that the question is about a government service and not an out-of-scope request.<br><br>← 6. **include::** RetrieveOfficialSources (UC-2).<br><br>← 7. **System** sends the question and the retrieved passages to the **AI Model Service**, which writes an answer using only those passages.<br><br>← 8. **System** checks that every fee, date, and document in the answer appears in a cited source. |

| | |
|---|---|
| | ← 9. **System** shows the answer in the question's language. Where available, it lists the steps in order, the required documents, fees and late fees, deadlines, and office hours.<br><br>← 10. **include::** ViewSourceCitation (UC-3).<br><br>← 11. **System** adds the question to the recent conversations list. |
| **Flow of Events for Extensions (Alternate Scenarios)** | 2a. User submits an empty input.<br><br>← 1. System ignores the submission and keeps the send button disabled.<br><br>5a. The question is not about a government service (for example, general knowledge or translation).<br><br>← 1. System tells the user that SuperAsk.AI only answers questions about Cambodian government services, and makes no retrieval or model call.<br><br>6a. No passage relevant enough to the question is found.<br><br>← 1. **extend::** DeclineUnsupportedQuestion (UC-4).<br><br>7a. The AI Model Service is unavailable or times out.<br><br>← 1. System falls back to showing the most relevant retrieved passage word for word, with its citation.<br><br>8a. The answer contains a claim that no cited source supports.<br><br>← 1. The system withholds the answer and tells the user it could not confirm every detail against an official source.<br><br>9a. The user asks a follow-up question about the answer.<br><br>→ 1. **extend::** AskFollowUpQuestion (UC-8). |

| Use Case UC-2: RetrieveOfficialSources | |
|---|---|
| **Related Requirements** | REQ-4, REQ-5, REQ-8 |
| **Initiating Actor** | Included by UC-1 AskQuestion (on behalf of Citizen / User) |
| **Actor's Goal** | To find the official passages that answer the question, so that the answer is based only on them. |
| **Participating Actors** | None (uses the system's stored official content) |
| **Preconditions** | A question has been submitted and passed the scope check. The content store contains official content collected by UC-11. |
| **Postconditions** | A ranked list of relevant passages is returned, each with its source metadata, or an empty result that triggers UC-4. |
| **Flow of Events for Main Success Scenario** | ← 1. **System** splits the Khmer or English question into search terms and adds known synonyms and abbreviations (for example, "MPWT" and "Ministry of Public Works and Transport").<br><br>← 2. **System** runs a keyword search and a meaning-based search over the stored content.<br><br>← 3. **System** merges and re-ranks the two result lists.<br><br>← 4. **System** removes outdated or replaced content, keeping only content that is currently in effect.<br><br>← 5. **System** checks that the top result scores above the minimum relevance threshold. |

| | |
|---|---|
| | ← 6. **System** returns the top passages, each with its ministry, source link, effective date, and last-verified date. |
| **Flow of Events for Extensions (Alternate Scenarios)** | 2a. The question is a follow-up that does not name its subject (for example, "how much is it?").<br><br>← 1. System rewrites the question using the earlier conversation before searching.<br><br>5a. No result reaches the minimum relevance threshold.<br><br>← 1. System returns an empty result, and the calling use case continues with DeclineUnsupportedQuestion (UC-4).<br><br>5b. The top result is related but does not mention most of what was asked.<br><br>← 1. System marks the result as low confidence, so the answer is shown with a note that the source may not fully answer the question. |

| **Use Case UC-3: ViewSourceCitation** | |
|---|---|
| **Related Requirements** | REQ-6, REQ-28, REQ-34 |
| **Initiating Actor** | Citizen / User |
| **Actor's Goal** | To confirm where the answer came from and how current it is. |

| | |
|---|---|
| **Participating Actors** | None |
| **Preconditions** | A sourced answer has been shown (UC-1). |
| **Postconditions** | The user has seen the source details and, if they chose to, opened the official source. |
| **Flow of Events for Main Success Scenario** | ← 1. **System** shows a "Verified Source" card under the answer, visually separate from the answer text.<br><br>← 2. **System** shows the ministry, document or page title, effective date, and last-verified date on the card.<br><br>→ 3. **User** reads the source details.<br><br>→ 4. **User** selects the link icon on the card.<br><br>← 5. **System** opens the official source page (ministry website, Facebook post, or Telegram post) in a new tab. |
| **Flow of Events for Extensions (Alternate Scenarios)** | 2a. The source's last-verified date is past its review date.<br><br>← 1. System marks the citation as possibly outdated and adds a note asking the user to confirm with the office before acting.<br><br>2b. The answer is supported by more than one source.<br><br>← 1. System shows one card per source, in the order they are cited.<br><br>3a. User finds the information incorrect.<br><br>→ 1. User selects "Report" on the answer and optionally adds a comment. |

| | |
|---|---|
| | ← 2. System records the report against the cited source for review. |

| **Use Case UC-7: GetOfficeContactAndLocation** | |
|---|---|
| **Related Requirements** | REQ-11, REQ-12, REQ-13, REQ-20, REQ-38, REQ-39 |
| **Initiating Actor** | Citizen / User |
| **Actor's Goal** | To know how to contact the responsible ministry or office and where to go in person. |
| **Participating Actors** | None (uses the stored office information) |
| **Preconditions** | The web interface is open. Contact and location details for the office have been collected from its official website, Facebook page, or Telegram channel (UC-11). |
| **Postconditions** | The user has the office's contact details and location, or a clear statement that no online contact is available together with the office location. |

| | |
|---|---|
| **Flow of Events for Main Success Scenario** | → 1. **User** asks for the contact and location of the office responsible for a procedure (for example, "can I have the contact and location of the ministry for this?").<br><br>← 2. **System** works out which procedure "this" refers to from the conversation (UC-8).<br><br>← 3. **System** finds the ministry or office responsible for that procedure.<br><br>← 4. **System** shows a contact card with the office name, address, office hours, phone number, and an "Open in Google Maps" link.<br><br>← 5. **System** shows the official source the contact details came from.<br><br>→ 6. **User** selects "Open in Google Maps".<br><br>← 7. **System** opens the office location in Google Maps in a new tab. |
| **Flow of Events for Extensions (Alternate Scenarios)** | 4a. No online contact information is available for the office.<br><br>← 1. System states "No verified online contact information is available."<br><br>← 2. System shows the office location and the Google Maps link instead.<br><br>4b. Neither contact nor location information is available from an official source.<br><br>← 1. System says it has no verified contact or location for this office and does not guess one.<br><br>3a. The procedure is handled by several offices (for example, a provincial office and the ministry in Phnom Penh).<br><br>← 1. System asks the user which province they are in. |

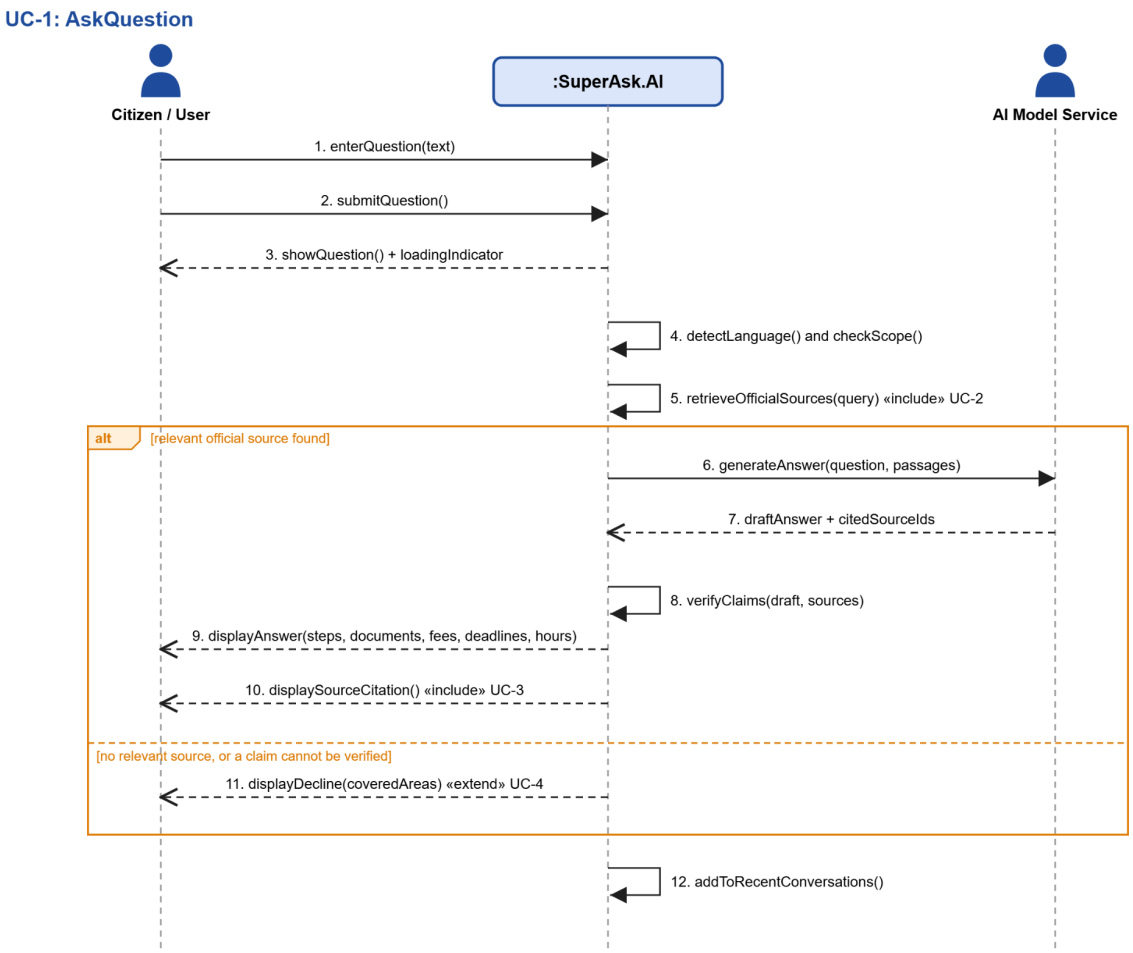
| | |
|---|---|
| | → 2. User answers, and the flow continues at step 4 with the matching office.<br><br>1a. User asks without naming or having discussed a procedure.<br><br>← 1. System asks which procedure or ministry the user needs.<br><br>→ 2. User names it, and the flow continues at step 3. |

| **Use Case UC-11: RefreshOfficialContent** | |
|---|---|
| **Related Requirements** | REQ-4, REQ-8, REQ-29, REQ-30 |
| **Initiating Actor** | Scheduler (System Timer) |
| **Actor's Goal** | To keep the stored government information current, so that answers and last-verified dates reflect what ministries have most recently published. |
| **Participating Actors** | Official Sources (Ministry Websites, Facebook Pages, Telegram Channels) |
| **Preconditions** | Each covered ministry's official website, public Facebook page, and public Telegram channel is registered in SuperAsk.AI's source list with its link and platform. |
| **Postconditions** | New or changed content is stored with its source link and fetch date, the last-verified date of every source checked successfully is updated, and a refresh report records which sources were checked, changed, or failed. |

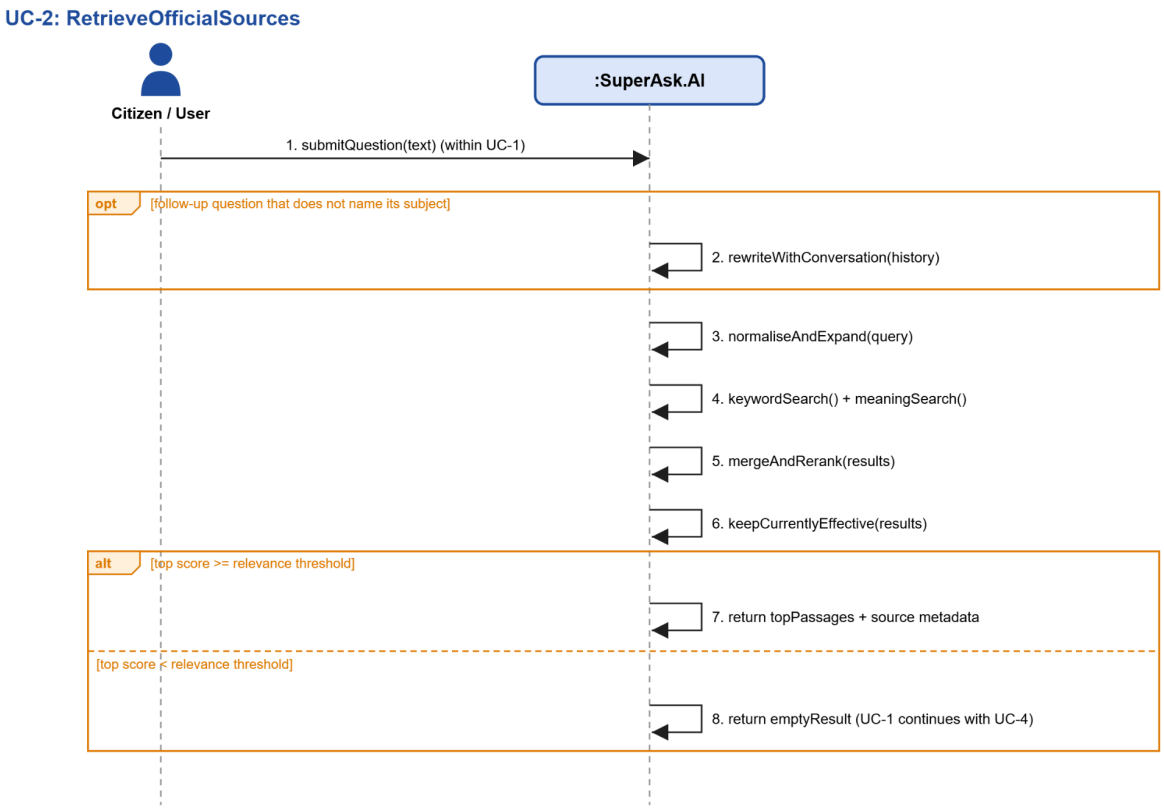
| | |
|---|---|
| **Flow of Events for Main Success Scenario** | → 1. Scheduler starts the content refresh at its set time.<br>← 2. System loads the list of registered official sources.<br>← 3. System requests the latest page or posts from each Official Source: a ministry web page, a public Facebook page, or a public Telegram channel.<br>→ 4. Official Source returns its content.<br>← 5. System extracts the text, removing menus and other repeated page elements.<br>← 6. System compares the text with the stored copy.<br>← 7. System saves any new or changed text with its source link and fetch date, and updates the information used for answers.<br>← 8. System updates the last-verified date of each source it checked successfully.<br>← 9. System records a refresh report listing the sources checked, changed, and failed. |
| **Flow of Events for Extensions (Alternate Scenarios)** | 2a. A new ministry's sources were added to the source list.<br>← 1. System includes them in this refresh, with no change to the answer-generation logic (REQ-30).<br>3a. The source's robots.txt rules do not allow automated access.<br>← 1. System skips the source and records it as blocked.<br>4a. The source does not respond or returns an error.<br>← 1. System keeps the stored copy and its previous last-verified date, and records the failure.<br>4b. The source returns a login or block page instead of real content.<br>← 1. System treats the response as a failure, not as changed content.<br>6a. The content has not changed.<br>← 1. System keeps the stored text and only updates the last-verified date. |

d. **System Sequence Diagrams**

The system sequence diagrams below show the messages exchanged between the actors and SuperAsk.AI, which is treated as a single black box, for the five use cases elaborated in Section 3c-iv. Solid arrows are requests, dashed arrows are responses, and the orange frames mark alternative (alt), optional (opt), and repeated (loop) parts of the flow.



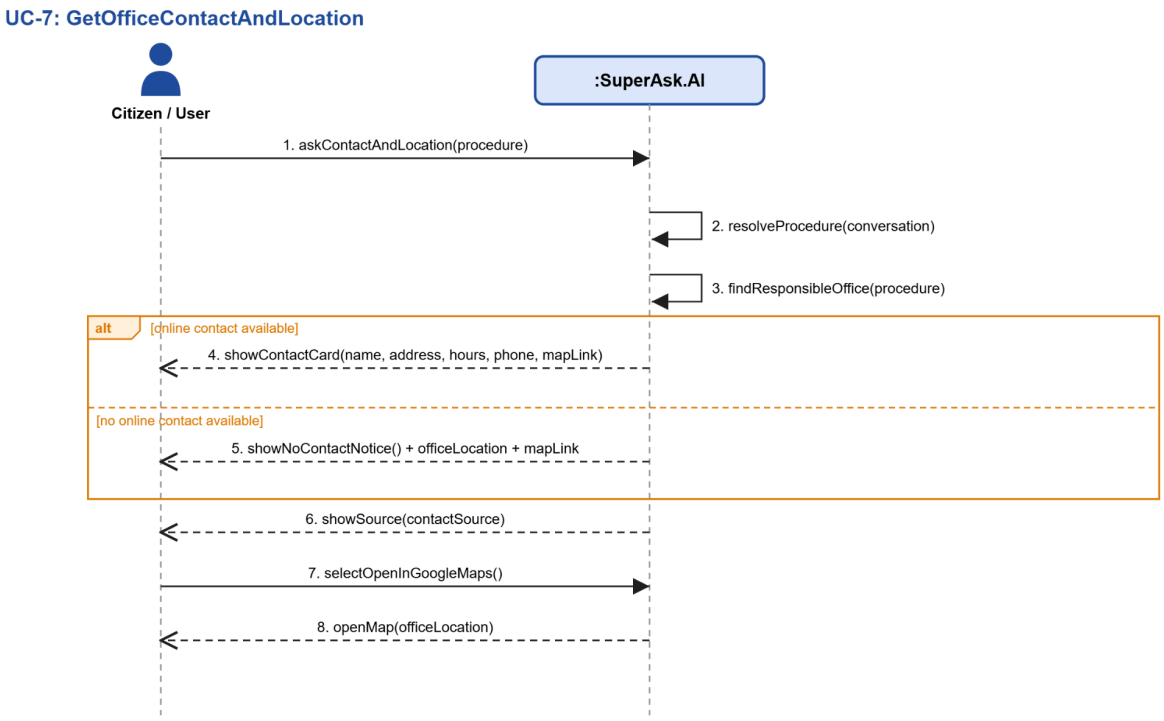

System sequence diagram for UC-1: AskQuestion



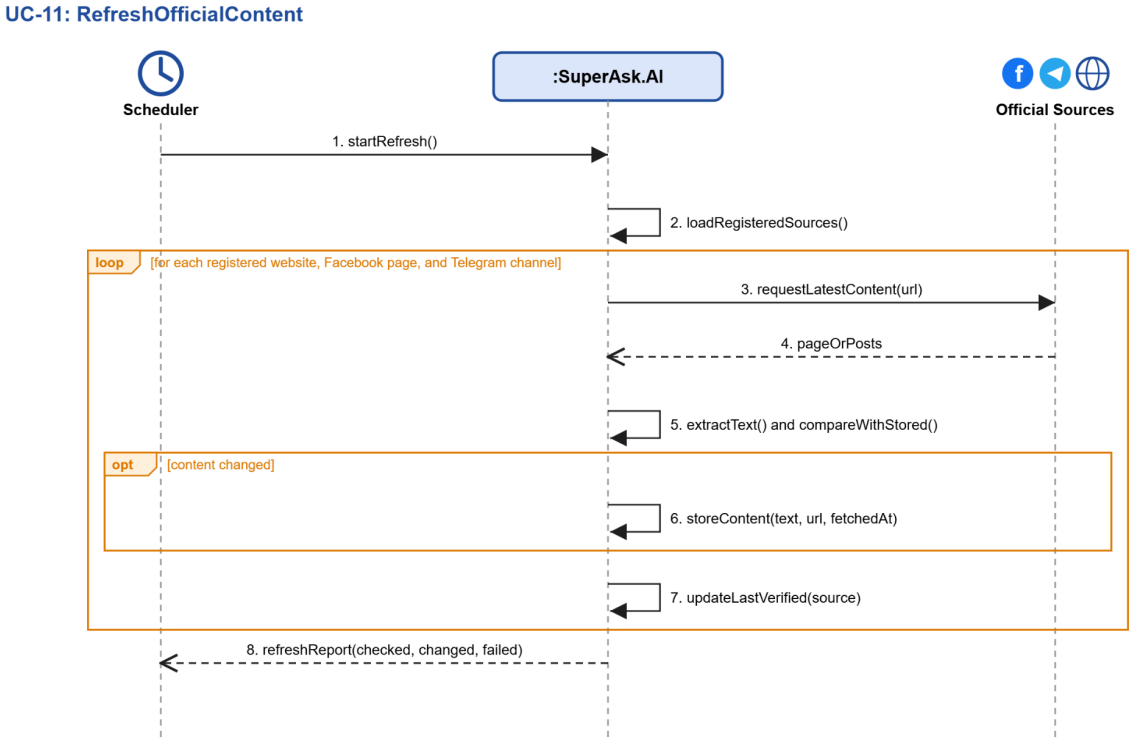

System sequence diagram for UC-2: RetrieveOfficialSources




System sequence diagram for UC-3: ViewSourceCitation




System sequence diagram for UC-7: GetOfficeContactAndLocation




System sequence diagram for UC-11: RefreshOfficialContent

## 4. User Interface Specification

### a. Preliminary Design

The SuperAsk.AI user interface is designed to provide citizens with a simple way to access information about Cambodian government services and procedures. The main interaction uses a chat-style interface so that users can ask questions in natural language rather than searching through multiple government websites or documents.

The interface design follows the requirements established in Report One. Users can enter and submit questions, view generated answers and verified official sources, select commonly requested procedures, return to recent questions, and use the interface in Khmer or English. The following mock-ups show the main navigation path, the information entered by the user, the controls used to complete each interaction, and the information displayed by SuperAsk.AI.

#### 1. Starting a Government-Service Question

The user can access the chat interface from the SuperAsk.AI Home Page by selecting Ask on the web or Ask now. The Home Page also provides access to language selection and supporting information about the service.

**Navigation Path:**
Home Page → Ask on the web / Ask now → Chat Interface → Enter Question or Select Procedure → Send / Select → Response

**User Interaction:**

1. The user opens the SuperAsk.AI Home Page.
2. If necessary, the user selects Khmer or English from the language control.
3. The user selects Ask on the web or Ask now to enter the chat interface.

**Primary controls:** Ask on the web, Ask now, language selection
**Result:** The user enters the SuperAsk.AI chat interface.

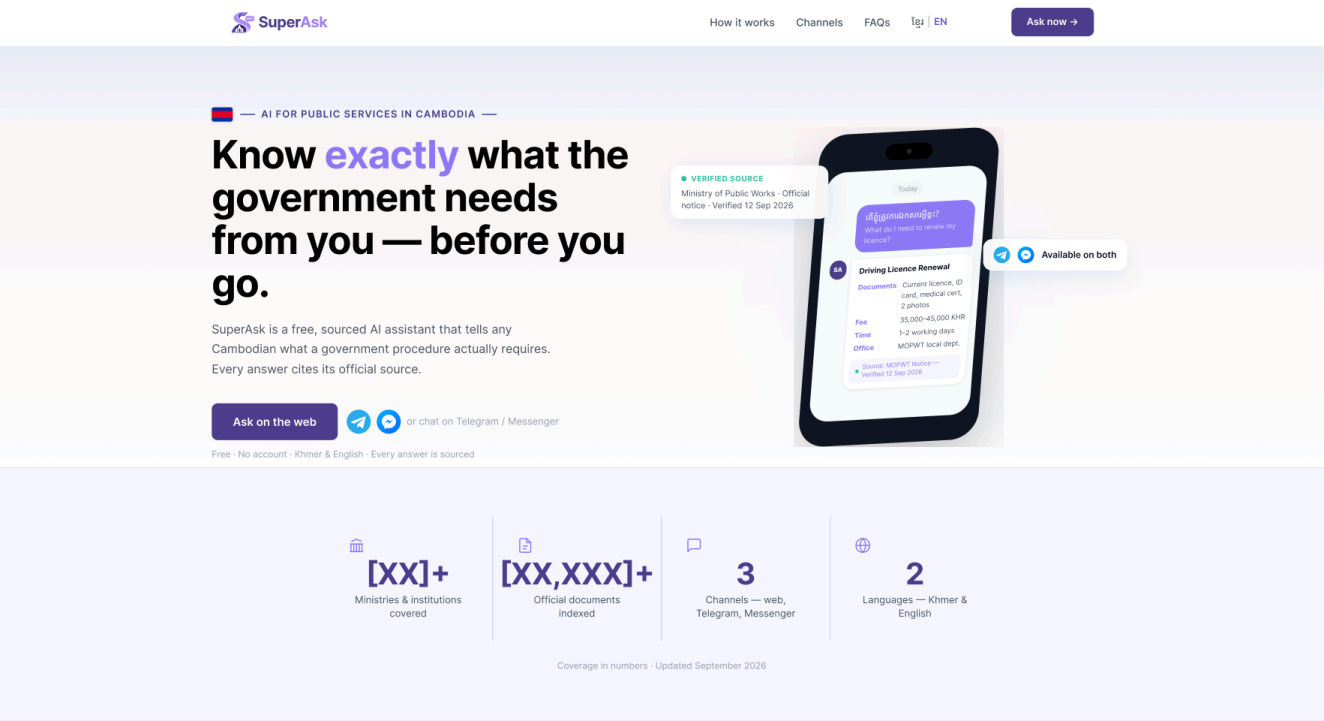



*Figure 1. SuperAsk.AI Home Page*

The initial chat screen provides three ways to begin an interaction. The user can enter a new question in the input field, select a previous question from Recent, or choose one of the Popular Procedures displayed in the interface.

4. The user may select a question from Recent, select a procedure from Popular Procedures, or use the main question input field.
5. To ask a new question, the user clicks the input field.
6. The user types a government-service question in Khmer or English.
7. The user presses the Send button.
8. SuperAsk.AI displays the corresponding response in the conversation area.
9. The user can continue asking follow-up questions through the same input field.

**Input field:** Government-service question

**Primary controls:** Send button, Recent questions, Popular Procedures, New chat, language control

**Displayed result:** The submitted question and corresponding response

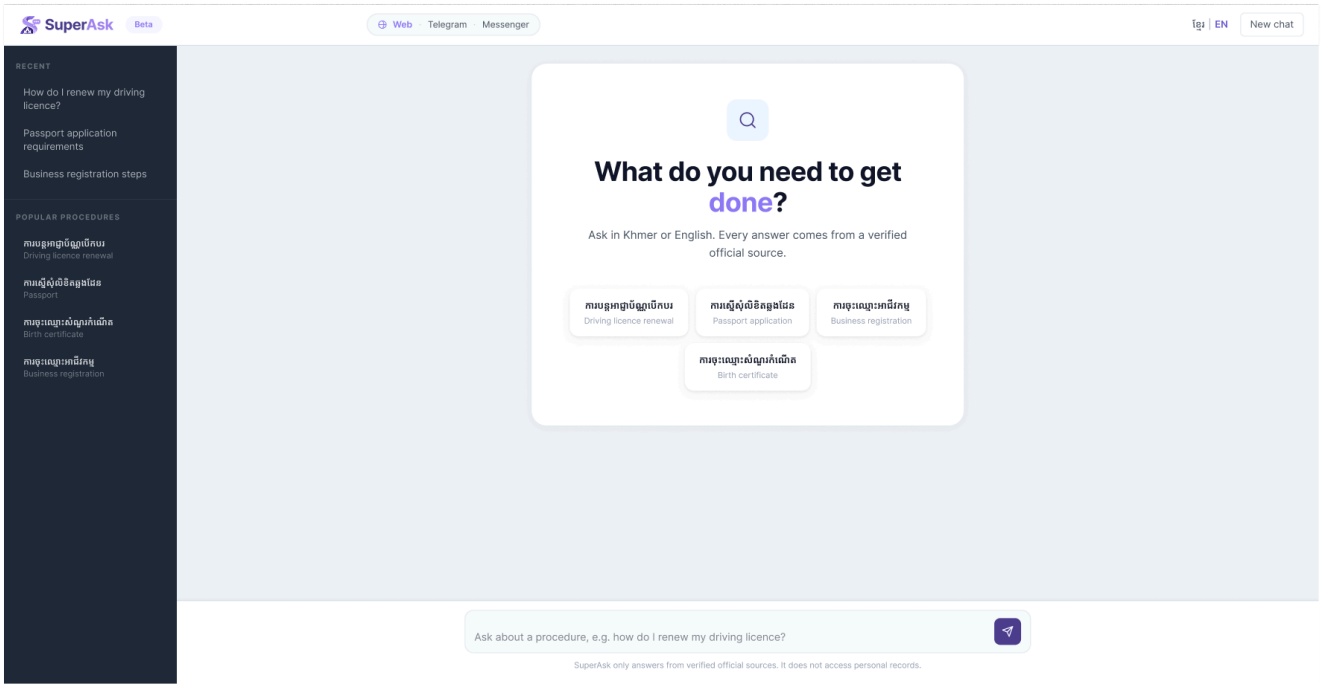



*Figure 2. SuperAsk.AI initial chat interface*

## 2. Viewing a Government-Procedure Answer

When SuperAsk.AI finds relevant information from a verified government source, the answer is displayed as a structured response card. The current prototype uses a driving licence renewal question to demonstrate this interaction.

The user's submitted question is displayed in a separate message bubble. The answer card then identifies the responsible government organization and presents the important procedure information in separate sections.

For the procedure shown in the mock-up, the response layout contains fields for Documents, Fee, Time, and Office, followed by the required procedure Steps. The verified source is displayed in a separate source area beneath the answer.

**Navigation Path:**
Chat Interface → Enter Question → Send → Procedure Answer → View Verified Source / Office Information

**User Interaction:**

1. The user enters a government-procedure question in the input field.
2. The user presses the Send button.
3. The submitted question appears in the conversation area.
4. SuperAsk.AI displays the relevant procedure answer.

5. The user reviews the required documents, fee, expected processing time, responsible office, and procedure steps when that information is available.
6. The user reviews the Verified Source displayed beneath the procedure information.
7. The user may select the source-link control to access the referenced source.
8. If location information is available, the user may select Show office on map.
9. The user may select Yes or No under Was this helpful? to provide feedback.
10. The user can enter a follow-up question through the input field or select New chat to begin another conversation.

**Input:** Government-procedure question
**Primary controls:** Send, verified-source link, Show office on map, Yes/No feedback, New chat
**Displayed result:** Procedure information, required documents, fees, processing time, responsible office, ordered steps, and verified source

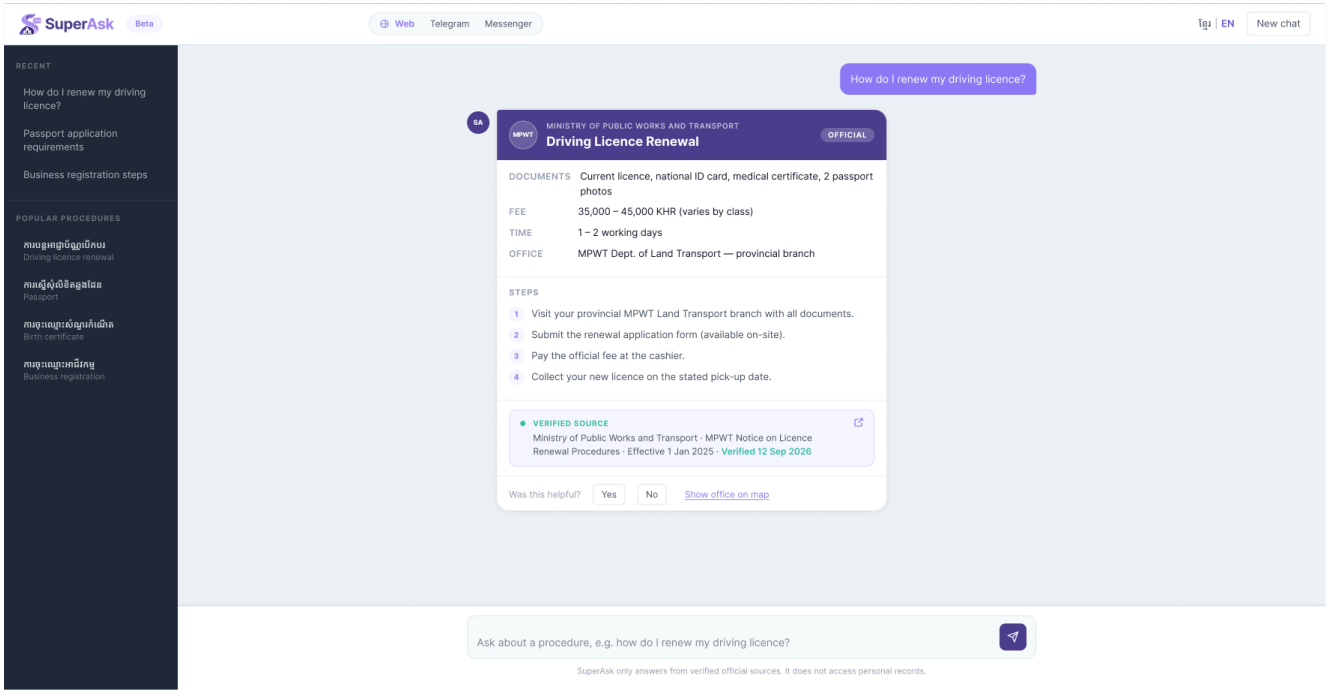



*Figure 3. SuperAsk.AI government-procedure answer with verified source.*

### 3. Handling an Unsupported or Out-of-Scope Question

SuperAsk.AI is specifically designed for questions related to Cambodian government procedures. Therefore, the interface also needs to clearly handle questions that fall outside the supported scope.

The out-of-scope mock-up demonstrates a user submitting the message *"hello."* Instead of generating an unrelated response, SuperAsk.AI displays a message explaining that the

service only answers questions about Cambodian government procedures and suggests relevant types of questions.

**Navigation Path:**
Chat Interface → Enter Unsupported Question → Send → Out-of-Scope Message → Enter Another Question

**User Interaction:**

1. The user types a question or message into the chat input field.
2. The user presses the Send button.
3. The submitted message appears in the conversation area.
4. If the request is outside the supported scope, SuperAsk.AI displays an out-of-scope response instead of presenting an unsupported answer.
5. The response explains that SuperAsk.AI is intended for Cambodian government procedures.
6. The interface suggests examples of supported topics, such as licences, certificates, registrations, or permits.
7. The user may return to the input field and enter another question.
8. Alternatively, the user may select New chat to begin a new conversation.

**Input:** Unsupported or unrelated question
**Primary controls:** Send button, question input, New chat
**Displayed result:** Out-of-scope guidance message

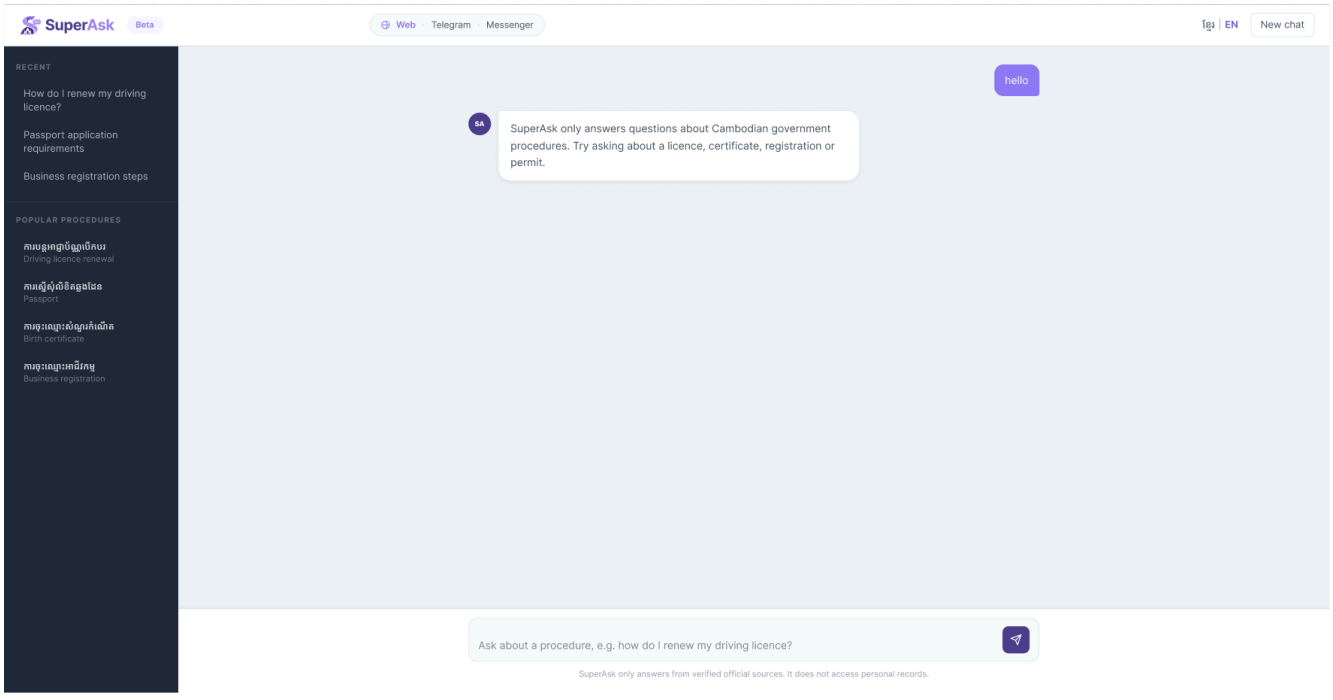



*Figure 4. SuperAsk.AI out-of-scope response.*

The interface was designed to minimize unnecessary navigation. Users can ask a question directly instead of first selecting a ministry or category, while popular procedures remain available as optional shortcuts. Procedure answers use a structured card rather than a long paragraph so that documents, fees, processing time, office information, steps, and verified sources are easier to identify.

## 5. Project Management

### Plan of Work

Following the submission of Report One, our team will continue developing the system through two main iterations. The first iteration will focus on developing and integrating the core functions needed for the first demonstration, including the web interface, bilingual question handling, retrieval from verified government sources, citation display, and guardrails for unsupported questions. The second iteration will focus on refinement, additional ministry coverage, testing, and improvements based on feedback from the first demonstration. The short-term activities are described in greater detail because they are the immediate priorities of the team. Later milestones may be adjusted based on technical progress, testing results, and instructor feedback.

| Period | Planned Work | Milestone |
|---|---|---|
| **Sep 21 - 25, 2026** | Complete customer requirements, business goals, functional/nonfunctional requirements, UI requirements, project management, and preliminary interface designs. | Report #1 Parts 1 & 2 |
| **Sep 26 - 30, 2026** | Review and revise requirements; ensure consistency between the problem statement, requirements, prototype, and team responsibilities. | Full Report #1 – Sep 30 |
| **Oct 1 - 6, 2026** | Continue collecting and verifying official government information. Develop interaction diagrams and begin implementing the retrieval and answer-generation workflow. | Report #2 Part 1 – Oct 6 |
| **Oct 7 - 12, 2026** | Develop class/system architecture diagrams. Continue development of the web interface, database, retrieval system, and bilingual processing. | Report #2 Part 2 – Oct 12 |
| **Oct 13 - 21, 2026** | Integrate the interface, retrieval pipeline, citation system, guardrails, and verified-source database. Conduct initial integration testing. | Full Report #2 – Oct 21 |
| **Oct 22 - Nov 1, 2026** | Implement the core use cases selected for the first demonstration. Test Khmer and English queries and verify responses against official sources. | First-demo implementation |

| | | |
|---|---|---|
| **Nov 2–8, 2026** | Debug the integrated system, test unsupported/out-of-scope questions, improve response quality, and prepare the demonstration. | First-demo preparation |
| **Nov 9, 2026** | Demonstrate the first working iteration and collect feedback. | Demo #1 |
| **Nov 10–16, 2026** | Apply feedback from Demo #1, expand verified government information, improve interface behavior, and conduct additional system testing. | Report #3 Part 1 – Nov 16 |
| **Nov 17–23, 2026** | Complete system testing and final documentation. Verify consistency between implementation, requirements, and reports. | Full Report #3 – Nov 23 |
| **Nov 24–Dec 3, 2026** | Refine the system, resolve remaining defects, improve ministry coverage, and prepare the final demonstration. | Final refinement |
| **Dec 4, 2026** | Present the completed SuperAsk.AI system. | Demo #2 |

The Gantt chart below summarizes the planned development activities and major project milestones from Report #1 through the final demonstration.




## Product Ownership and Team Responsibilities

SuperAsk.AI is being developed collaboratively by six team members. Each member has a primary project role based on their technical and organizational strengths, while major design, testing, and implementation decisions are discussed collectively. The Project Lead coordinates work between the different roles and monitors progress against the semester schedule.

| Team Member | Role | Work Completed / So Far | Current Responsibilities | Future Responsibilities |
|---|---|---|---|---|
| **Yean Fong** | **Project Lead** | Coordinated the project proposal, project scope, team organization, scheduling, and requirements preparation. Contributed to design and implementation documentation. | Coordinate Report #1, requirements, deadlines, communication between team members, and overall project consistency. | Monitor milestones, coordinate system integration and demos, assist testing/documentation, and manage final project delivery. |
| **Phallamonunhh Chun** | **AI/ML Engineer** | Contributed to business analysis, AI planning, cybersecurity considerations, and the SuperAsk.AI concept. | Develop and evaluate the RAG pipeline, model integration, retrieval behavior, and AI guardrails. | Improve answer accuracy, retrieval performance, refusal behavior, and bilingual AI handling. |
| **Bunkheang Heng** | **Web/API Developer** | Contributed technical development, system architecture planning, full-stack concepts, and database planning. | Develop the web application, APIs, database connections, and integration with the AI system. | Complete integration, optimize backend functions, fix technical defects, and support deployment. |
| **Doungkhe Neang** | **Data/Content Curator** | Contributed data organization, content planning, documentation, and source research. | Collect, organize, verify, and categorize official government content used by the system. | Expand ministry coverage, maintain source metadata, and help ensure information remains current and properly cited. |

| | | | | |
|---|---|---|---|---|
| **Sereyprophea Sarin** | **QA/Testing Lead** | Contributed documentation, communication, interface discussion, and initial testing planning. | Develop test questions and check accuracy, bilingual responses, source citation, and refusal behavior. | Conduct integration and user testing, record defects, collect user feedback, and verify corrections before demonstrations. |
| **Heejae Choi** | **UX/UI Designer** | Contributed interface planning, UX/UI concepts, page layout, and prototype development. | Refine user flows, chat interface, responsive layouts, and presentation of answers and citations. | Improve the interface based on user testing and ensure consistency across desktop/mobile views and supported channels. |

### Management and Coordination

The Project Lead will maintain the overall project schedule and coordinate dependencies between the sub-teams. Development tasks will be reviewed regularly to identify delays or technical problems early. Team members remain primarily responsible for their assigned areas, but implementation and testing will overlap when integration requires cooperation between AI, content, interface, backend, and QA work.

| Team Member | Competencies | Assigned Area |
|---|---|---|
| **Yean Fong** | Project management, organizational skills, assertive communication, Python, C++, JavaScript, HTML, CSS, Design | **Management / Integration** |
| **Phallamonunhh Chun** | UX/UI, Business Analysis, Cybersecurity, CSS, C++, AI | **AI / Data** |
| **Bunkheang Heng** | Technical development, System architecture, Full Stack Development, Database Design, Machine Learning | **Development** |

| | | |
|---|---|---|
| **Doungkhe Neang** | UX/UI, data curation and management, technical documentation, information organization | **AI / Data** |
| **Heejae Choi** | UX/UI, interface planning, C++, JavaScript, HTML, CSS, documentation | **UX / UI** |
| **SereyProphea Sarin** | Documentation, communication, UX/UI, Java, JavaScript, HTML, CSS | **Testing / QA** |

## References


**SuperAsk.AI Team AO3. (2026).** ***SuperAsk.AI GitHub repository*****.**
https://github.com/Monunhh/Super-ask.git
Used for project source code, version control, and team development.

**Open Government Products. (2026).** ***AskGov Singapore*****. Singapore Government.**
https://ask.gov.sg/
Used as a reference for how government information can be organized by agency and presented through a searchable question-and-answer interface. AskGov lists government organizations and public-service questions in one place.